PROFITABILITY ANALYSIS OF NASSAU CANDY DISTRIBUTOR USING PYTHON AND POWER BI

Research Paper

Project Title: Profitability Analysis of Nassau Candy Distributor Using Python and Power BI

Submitted By: Arshi Zainab

Project Organization: Unified Mentor

Tools and Technologies: Python, Pandas, Data Analysis, Jupyter Notebook, Microsoft Power BI

ABSTRACT

Profitability analysis plays an important role in understanding the financial performance of a business. Companies need to know how much revenue they generate, how much they spend, and which products or business divisions contribute most to their profits. Without a clear understanding of these factors, it can be difficult to identify areas that require improvement.

This project focuses on analysing the profitability of Nassau Candy Distributor using Python and Microsoft Power BI. The main objective is to examine sales, costs, gross profit, gross margin, product performance, and division-level performance. Python is used for data preparation and analysis, while Power BI is used to present the results through an interactive dashboard.

The dashboard includes key performance indicators such as total sales, total gross profit, total cost, and gross margin percentage. It also contains visualisations for sales by division, gross profit by division, product profitability segments, product-level performance, and sales trends over time.

The dashboard displays total sales of approximately 141.78K, total gross profit of 93.44K, total cost of 48.34K, and an overall gross margin of 65.90%. These figures provide a useful overview of the financial performance represented in the analysed dataset.

The project demonstrates how data analysis and visualisation can help organise financial information, identify performance patterns, and support business decision-making. It also highlights the importance of using accurate data and appropriate visualisations when evaluating business profitability.

Keywords: Profitability Analysis, Sales Analysis, Gross Profit, Gross Margin, Python, Power BI, Data Visualisation, Business Intelligence.

---

1. INTRODUCTION

In the modern business environment, data plays an important role in planning and decision-making. Businesses generate large amounts of information through their daily operations, including sales transactions, product details, costs, and revenue records. Analysing this information can help organisations understand their performance and identify opportunities for improvement.

Profitability is one of the key measures used to evaluate business performance. A business may generate high sales but still experience lower profits if its operating costs are high. Therefore, examining sales figures alone does not provide a complete picture of financial performance. It is also necessary to analyse costs, gross profit, and profit margins.

Nassau Candy Distributor is the focus of this project. The project uses data analysis techniques to examine the financial performance represented in the available dataset. Particular attention is given to product-level profitability, division-level sales, gross profit, and changes in sales over time.

Python is used to prepare and analyse the data, while Microsoft Power BI is used to create a visual dashboard. The dashboard brings important financial measures together in one place, making it easier to review the results without manually examining every record.

The purpose of this project is not only to calculate financial indicators but also to present them in a clear and understandable format. By combining data analysis with interactive visualisation, the project demonstrates a practical approach to business performance analysis.

2. PROBLEM STATEMENT

Businesses need reliable information about their sales, costs, and profits to make informed decisions. However, when this information is stored in large datasets, identifying important patterns can take time. Looking at individual transaction records may not provide an immediate understanding of which products perform well or how profitability differs across business divisions.

The main problem addressed by this project is the need for a clear and structured analysis of Nassau Candy Distributor's financial data.

The project aims to organise the available information and present important performance indicators through a Power BI dashboard. It focuses on understanding sales performance, comparing gross profit across products and divisions, examining profitability segments, and reviewing sales trends.

A visual representation of these measures can make the results easier to understand and help users identify areas that may require further investigation.

3. OBJECTIVES OF THE PROJECT

The main objectives of this project are:

1. To analyse the sales data available for Nassau Candy Distributor.
2. To calculate and examine total sales, total cost, and total gross profit.
3. To understand overall gross margin performance.
4. To compare sales performance across the divisions represented in the dataset.
5. To analyse gross profit at the product level.
6. To classify products into profitability segments for comparison.
7. To examine sales trends using the available order date information.
8. To develop an interactive Power BI dashboard for presenting the results.
9. To make financial information easier to interpret through charts, KPI cards, and slicers.
10. To demonstrate how data analysis tools can support business reporting and decision-making.

4. SCOPE OF THE STUDY

The scope of this project is limited to the data available for analysis. It covers financial performance indicators and product-level and division-level comparisons represented in the dataset.

The study focuses on the following areas:

- Sales performance
- Gross profit analysis
- Cost analysis
- Gross margin percentage
- Product-level profitability
- Division-level performance
- Profitability segment comparison
- Sales trends over time

Python is used for data preparation and analysis, and Power BI is used to present the results visually.

The project provides a descriptive view of the available data. Its findings can help identify patterns and areas for further investigation, but they should not be interpreted as proof of the causes behind those patterns. The analysis also does not establish future sales forecasts or the financial impact of changing product allocations.

5. TOOLS AND TECHNOLOGIES USED

5.1 Python

Python is a programming language widely used for data analysis. In this project, it is used to work with the dataset and prepare the information required for analysis.

Python helps make data processing more organised and repeatable. It is also useful for examining numerical values and preparing data for visualisation.

5.2 Pandas

Pandas is a Python library used to manipulate and analyse structured data. It provides DataFrame structures that make it easier to work with rows and columns.

In this project, Pandas is used as part of the data preparation and analysis process. It can be used to inspect columns, examine records, handle data formatting, and prepare the final dataset for use in Power BI.

5.3 Jupyter Notebook

Jupyter Notebook provides an interactive environment in which Python code can be executed step by step. It allows code, outputs, and explanations to be kept together.

It is useful for documenting the data analysis workflow and checking the results produced during different stages of the project.

5.4 Microsoft Power BI

Microsoft Power BI is a business intelligence tool used to create reports and dashboards from data.

In this project, Power BI is used to present financial indicators through KPI cards, charts, and slicers. These visuals help users compare performance across products and divisions and examine the available sales trends.

5.5 CSV

CSV, or Comma-Separated Values, is a common file format for storing tabular data. It allows data prepared in Python to be transferred into Power BI.

The prepared dataset was exported in CSV format for use in the dashboard development process.

6. DATASET DESCRIPTION

The project uses the available Nassau Candy dataset for analysing sales and profitability.

The fields used in the dashboard include the following:

Sales: Represents the sales value recorded in the dataset.

Gross Profit: Represents the gross profit value associated with the relevant records.

Cost: Represents the cost values included in the dataset.

Gross Margin (%): Represents the gross margin percentage used to evaluate profitability relative to sales.

Product ID: Provides an identifier for a product.

Product Name: Identifies products and allows their performance to be compared.

Division: Identifies the division associated with a record and supports division-level analysis.

Order Date: Provides date information used to examine sales over time.

The exact interpretation of each field depends on the definitions and structure of the original dataset.

Before analysis, it is important to inspect the available columns, check data types, and ensure that the fields used in calculations contain appropriate values. This helps reduce the risk of misleading results.

7. METHODOLOGY

The project follows a data analysis workflow that moves from data preparation to financial analysis and dashboard development.

7.1 Data Loading

The dataset was loaded into the Python environment for inspection and analysis. The initial stage involved examining the structure of the data, including its columns and records.

Understanding the dataset at the beginning is important because the accuracy of later calculations depends on the quality and structure of the input data.

7.2 Data Inspection and Preparation

The data was examined to understand the available variables and identify the fields required for profitability analysis.

Data preparation generally involves checking column names, reviewing data types, examining missing values, and ensuring that numerical fields can be used in calculations.

Only the information required for the intended analysis should be used in the final reporting dataset.

7.3 Financial Analysis

The financial analysis focuses on sales, gross profit, cost, and gross margin.

These measures provide different perspectives on business performance. Sales show the value of sales recorded in the dataset, gross profit indicates the profit remaining after the relevant costs represented in that measure, and gross margin expresses gross profit as a proportion of sales.

Examining these indicators together provides a more useful view than considering sales alone.

7.4 Product-Level Analysis

Product-level analysis is used to compare the performance of individual products.

The dashboard includes gross profit by product and sales by product. These charts help users identify differences in the financial contribution of products.

Products with higher sales do not necessarily have the highest gross profit. Comparing both measures can therefore provide a more balanced understanding of product performance.

7.5 Division-Level Analysis

Division-level analysis compares sales and gross profit across the divisions available in the dataset.

The dashboard includes separate visualisations for sales by division and gross profit by division. This allows users to compare revenue generation and gross profit contribution.

The usefulness of this comparison depends on the number and variety of divisions actually represented in the dataset.

7.6 Profitability Segmentation

The dashboard includes a visualisation showing products by profitability segment.

Segmentation groups products into categories to make comparisons easier. The segment labels used in this project include High Profitability, Medium Profitability, and Low Profitability.

In the dashboard calculation, gross margin thresholds are used to assign these labels. A margin of 60% or above is classified as High Profitability, a margin of at least 30% but below 60% is classified as Medium Profitability, and a margin below 30% is classified as Low Profitability.

These thresholds are analytical choices used for grouping the data. They should not be treated as official Nassau Candy business targets unless independently confirmed.

Segmentation provides a simplified way to review product performance. It does not replace a detailed examination of sales volume, costs, or other business considerations.

7.7 Dashboard Development

After preparing the data, the CSV dataset was imported into Power BI Desktop.

KPI cards were created to display the main financial indicators. Additional charts were added to compare sales, gross profit, product profitability, and sales trends over time.

Slicers were included to allow users to filter the dashboard by division and product name. This makes the report more interactive and allows users to focus on selected parts of the dataset.

The final dashboard brings several related measures together on a single report page.

8. FINANCIAL MEASURES USED

8.1 Total Sales

Total sales represent the sum of the sales values included in the analysed records.

Formula:

Total Sales = Sum of Sales

This measure provides an overview of the total sales value represented in the dataset.

8.2 Total Gross Profit

Total gross profit represents the sum of the gross profit values included in the dataset.

Formula:

Total Gross Profit = Sum of Gross Profit

This measure helps users understand the total gross profit represented by the analysed records.

8.3 Total Cost

Total cost represents the sum of the cost values included in the dataset.

Formula:

Total Cost = Sum of Cost

This measure provides an overview of the recorded costs included in the analysis. Its precise accounting interpretation depends on how the original dataset defines the Cost field.

8.4 Gross Margin Percentage

Gross margin percentage expresses gross profit as a percentage of sales.

Formula:

Gross Margin (%) = (Total Gross Profit / Total Sales) × 100

The measure helps compare gross profit with the sales value that generated it.

For the overall margin, total gross profit is divided by total sales. This is different from simply taking the average of individual row-level margin percentages.

8.5 Profitability Segments

Products are grouped into profitability segments using the gross margin percentage.

The classification applied in the dashboard is:

- High Profitability: Gross margin greater than or equal to 60%.
- Medium Profitability: Gross margin greater than or equal to 30% and below 60%.

- Low Profitability: Gross margin below 30%.

This classification makes it easier to compare products across broad margin categories.

9. POWER BI DASHBOARD DESIGN

The Power BI dashboard was developed to provide a visual overview of the financial information.

9.1 KPI Cards

Four KPI cards display the main financial measures:

1. Total Sales
2. Total Gross Profit
3. Total Cost
4. Gross Margin Percentage

These cards provide a quick summary of the financial results and allow users to understand the overall figures before examining individual charts.

9.2 Sales by Division

The Sales by Division chart compares the sales values associated with the divisions in the dataset.

It helps users examine how sales are distributed across the available divisions.

9.3 Gross Profit by Division

The Gross Profit by Division chart compares gross profit across divisions.

Considering this chart alongside Sales by Division helps distinguish sales performance from gross profit performance.

9.4 Gross Profit by Product

The Gross Profit by Product chart compares the gross profit associated with individual products.

It helps identify products with relatively higher or lower gross profit values in the analysed dataset.

9.5 Products by Profitability Segment

This visualisation groups products according to the profitability categories defined in the analysis.

It provides a simple overview of how products are distributed across the selected margin thresholds.

9.6 Monthly Sales Trend

The sales trend visualisation uses the available order date information to examine changes in sales over time.

A monthly view is useful for examining shorter-term patterns that may not be visible in an annual summary.

For a clear monthly trend, the horizontal axis should represent the date hierarchy at the appropriate year and month level, and the vertical axis should represent the sum of sales. Order IDs are

transaction identifiers rather than calendar periods and should not be used as a substitute for the date axis in a monthly trend chart.

9.7 Sales by Product

The Sales by Product chart compares sales values across products.

This visualisation complements the gross profit chart because products with higher sales may not necessarily have the highest gross profit.

9.8 Division and Product Slicers

The dashboard includes slicers for Division and Product Name.

These controls allow users to select a division or product and examine the relevant results. Slicers make the dashboard more interactive, provided that the underlying data and visual interactions are configured correctly.

The values available in a slicer depend on the dataset and the filters applied to the report.

10. RESULTS AND DISCUSSION

The Power BI dashboard displays the following financial figures for the analysed data:

Financial Measure	Dashboard Result
Total Sales	141.78K
Total Gross Profit	93.44K
Total Cost	48.34K
Gross Margin Percentage	65.90%

Note: The values above are reported in the format displayed on the dashboard. The currency and units should be confirmed from the original dataset.

10.1 Sales Performance

The dashboard reports total sales of approximately 141.78K.

This figure provides an overall view of the sales value represented in the analysed records. However, total sales alone cannot explain whether the business is operating profitably.

For that reason, sales are examined alongside gross profit and cost.

10.2 Gross Profit Performance

The total gross profit displayed on the dashboard is approximately 93.44K.

This measure provides an overview of the gross profit recorded in the dataset. The product-level and division-level charts provide additional detail by showing how gross profit differs between the available categories.

These comparisons can help identify products or divisions that deserve further examination. The reasons behind differences in performance would require additional investigation into factors such as sales volume, prices, costs, and product characteristics.

10.3 Cost Analysis

The total cost displayed on the dashboard is approximately 48.34K.

Examining cost alongside sales and gross profit helps provide context for the financial results.

However, the interpretation of this figure depends on the definition of Cost in the original dataset. It should not automatically be treated as a complete measure of every business expense.

10.4 Gross Margin Analysis

The dashboard displays an overall gross margin of 65.90%.

Gross margin indicates the proportion of sales represented by gross profit. Using the reported figures, the overall margin is approximately 65.9%, subject to rounding in the dashboard.

This is an aggregate measure calculated from total gross profit divided by total sales. It does not mean that every product has a gross margin of 65.90%.

Individual products may have higher or lower margins, which is why the product profitability segmentation is useful.

10.5 Product-Level Performance

The dashboard includes sales and gross profit comparisons at the product level.

These visuals make it possible to compare the relative financial contribution of products. They can help users identify products that generate higher gross profit and products that may require further investigation.

A product with high sales but comparatively low gross profit may warrant a closer review of its costs or pricing. Similarly, a product with a high gross margin but low sales may contribute less to total gross profit than its margin percentage alone suggests.

Such observations should be confirmed by examining the underlying product records before making business decisions.

10.6 Division-Level Performance

The Sales by Division and Gross Profit by Division charts provide two different views of division-level performance.

When multiple divisions are present, these charts can help users compare their relative contributions. If the current slicer displays only one division, the available data and report filters should be checked before drawing conclusions about differences between divisions.

The division-level analysis is therefore most useful when all relevant division records are included and the slicers are configured correctly.

10.7 Sales Trends

The date-based sales visualisation is intended to show how sales change over time.

A monthly trend can help users identify periods with relatively higher or lower sales. However, a meaningful interpretation requires a correctly configured date axis and appropriate aggregation of sales.

The dashboard should be checked to ensure that the monthly chart displays sales by year and month rather than grouping the data by individual order identifiers.

10.8 Overall Interpretation

The dashboard brings the main financial indicators and product-level comparisons together in a single report.

The reported figures indicate that the analysed records contain approximately 141.78K in sales and 93.44K in gross profit, with an overall gross margin of 65.90%.

The additional charts provide ways to investigate how performance varies by product, division, and time period. These views are valuable because an overall total does not show the distribution of performance across individual products.

The results describe the data included in the analysis. They should not be generalised to the entire business without confirming the dataset's coverage, date range, and completeness.

11. BUSINESS VALUE AND PRACTICAL APPLICATIONS

The project demonstrates several ways in which data analysis can support business reporting.

Improved visibility: The KPI cards bring the main financial indicators together, reducing the need to examine